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2024_28_III_OOPS Using Java Lab

2028_REC_OOPS using Java_Week 6_Q2

Attempt : 1
Total Mark : 10
Marks Obtained : 10

Section 1 : Coding

1. Problem Statement

Alice is managing an online store and wants to implement a program using inheritance to calculate the selling price of products after applying discounts.

Guide her by following the instructions:

Create a base class called Product with a public double attribute price.Create a subclass called DiscountedProduct, which extends Product and includes a private double attribute discount rate. This subclass has a method called calculateSellingPrice() to determine the final selling price after applying the discount.

Formula: Discounted selling price = price * (1 - discount rate)

Input Format

The first line of input consists of a double value p, the initial price of the product.

The second line consists of a double value d, the discount rate.

Output Format

The output prints "Rs. X", where X is a double value, representing the calculated discounted selling price, rounded off to two decimal places.

If the discount rate is greater than 1, print "Not applicable".

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 50.00
0.20

Output: Rs. 40.00

Answer

```
import java.util.Scanner;

// You are using Java


/**
 * The base class representing a product with a price.
 */
class Product {
   // Public attribute to store the price of the product.
   public double price;
}

/**
 * A subclass representing a product with a discount.
 * It inherits the price from the Product class.
 */
class DiscountedProduct extends Product {
   // Private attribute to store the discount rate (e.g., 0.20 for 20%).
   private double discountRate;
```

```
    /**
     * Constructor for DiscountedProduct.
     * @param price The initial price of the product, inherited from the parent.
     * @param discountRate The discount rate to be applied.
     */
    public DiscountedProduct(double price, double discountRate) {
        this.price = price; // Set the price in the parent class
        this.discountRate = discountRate;
    }

    /**
     * Calculates the final selling price after applying the discount.
     * @return The discounted selling price.
     */
    public double calculateSellingPrice() {
        return price * (1 - discountRate);
    }
}


class ProductPricing {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);

        double initialPrice = scanner.nextDouble();
        double discountRate = scanner.nextDouble();
        DiscountedProduct discountedProduct = new
DiscountedProduct(initialPrice, discountRate);
        double sellingPrice = discountedProduct.calculateSellingPrice();

        if (sellingPrice >= 0) {
            System.out.printf("Rs. %.2f%n", sellingPrice);
        } else {
            System.out.println("Not applicable");
        }
        scanner.close();
    }
}
```

Status : Correct ***Marks : 10/10***